How the Brain Learns

Glossary

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Each entry gives the meaning a term has in this book and the chapter where it is introduced or most fully explained.

Acetylcholine
A neuromodulator released mainly from the basal forebrain, including the nucleus basalis of Meynert, and from brainstem nuclei; it is also the fast transmitter at the junction between nerve and muscle. High levels, as in attentive waking, favor encoding new information, while low levels, as in slow-wave sleep, favor recall and replay. (Chapter 3)

Action potential (spike)
A brief, all-or-none electrical pulse, about a millisecond long, by which a neuron signals. Voltage-gated sodium channels open and drive the membrane positive, then close as potassium channels return it to rest; the pulse starts at the axon initial segment, regenerates itself along the axon, and also spreads back into the dendrites. Information is carried by its rate and timing, not its size. (Chapter 2)

Active inference
In the free-energy framework, reducing prediction error by acting to change sensations so that they match the model, rather than by changing the model. Exploration appears in it as an assumed "epistemic" part of the free energy expected in the future. (Chapter 12)

Adaptive learning rate
A learning rate that rises when the learner is uncertain or the world has likely changed and falls as a new estimate is confirmed, as in Kalman's solution. People's learning rates, and their pupil size, behave this way; it is the clearest candidate for what all the brain's error signals have in common. (Chapter 12)

Adenosine triphosphate (ATP)
The small molecule that cells use as an energy currency. Splitting it into adenosine diphosphate and phosphate releases free energy that pays for nearly every energy-requiring process, from pumping ions to building proteins; mitochondria make most of it. (Chapter 0)

Alliesthesia
The change in a stimulus's pleasantness with the body's state, as when a sweet taste that pleases a fasting person becomes unpleasant after a glucose drink. It shows that pleasure reflects a stimulus's usefulness to the body at that moment rather than a fixed property of the stimulus. (Chapter 7)

Allostasis

Literally “stability through change”: regulation by prediction, in which the brain anticipates what the body will need and adjusts its physiology in advance instead of waiting for an error. Critics regard it as homeostasis with good forecasting rather than a new principle. (Chapter 7)

Allostatic load

The cumulative cost of chronically activated stress responses. In rodents, chronic stress shrinks dendrites in the hippocampus, mostly reversibly, and enlarges them in the amygdala, more persistently. (Chapter 7)

Ambiguity aversion

A dislike of unknown odds as such, shown when people prefer betting on an urn of known composition over one whose mix is unknown, whichever color they bet on. It may make sense for an animal whose ignorance often means that someone else knows more. (Chapter 8)

Amygdala

A cluster of nuclei deep in the temporal lobe, named for its almond shape, that learns the positive and negative value of cues. It is needed for learned fear of external threats and for the emotional enhancement of memory, but it is not simply a fear center. (Chapter 7)

Analog-digital hybrid computation

The proposal, which the book describes as roughly how nervous systems are built, that noisy components compute most efficiently in graded form over short distances, where graded signals are cheap and rich, and restore the signal to discrete form, such as spikes, before noise accumulates over long distances. (Chapter 13)

Anterior forebrain pathway (Area X)

In songbirds, a side loop of the song system that runs through Area X, part of the bird’s basal ganglia, and back onto the motor pathway that produces song. It injects variability into the song, and dopamine signals about whether a rendition sounded better or worse than expected bias the variation toward better versions. (Chapter 9)

Astrocyte

A type of glial cell. Activating astrocytes in the mouse hippocampus has been reported to induce long-lasting potentiation and enhance memory, but other laboratories found that astrocyte signaling did not affect plasticity, so whether they take part in learning is disputed. (Chapter 15)

Automaticity

Performance that is fast, consistent, and cheap in attention, and for the same reasons hard to inspect and hard to change. It develops only when the mapping from situation to response is consistent, and once developed it is hard to suppress. (Chapter 9)

Axon

A long, slender projection of a neuron specialized for transmitting action potentials to other neurons or target cells. (Chapter 2)

Axon initial segment

The first few tens of micrometers of the axon, packed with voltage-gated sodium channels at far higher density than the soma or dendrites. It is where the summed input is compared to threshold and the action potential begins. (Chapter 2)

Back-propagating action potential

An action potential that, after starting in the axon, spreads actively back into the dendritic tree. It reports to the neuron's synapses that the cell has just fired, a key signal for synaptic learning. (Chapter 2)

Backpropagation

The algorithm by which artificial networks carry the error at the output backward, layer by layer, and give every weight an update proportional to its exact contribution to that error. It requires a complete, labeled, differentiable description of the system; whether the brain approximates it is an open question. (Chapter 3)

Basal ganglia

A set of subcortical nuclei that select one action from the many the cortex proposes. Their output holds motor circuits under constant inhibition, and a cortical pattern acting through the striatum releases a single action by disinhibition; dopamine reaching the striatum is where reward prediction error meets action selection. (Chapter 4)

Behavior

Observable actions or responses of an organism to internal or external conditions.

Bit

The unit of information: the amount needed to decide between two equally likely possibilities. (Chapter 13)

Bitter lesson

Sutton's 2019 argument that, over 70 years of artificial intelligence research, general methods that make use of more computation have always beaten methods built on human knowledge of the problem. Any principle borrowed from the brain must pass this test at scale. (Chapter 10)

Bounded rationality

Herbert Simon's view that real choosers satisfice, searching until they find an option good enough rather than the best, because search itself is costly. Its modern extension, resource rationality, treats many apparent biases as optimal once the cost of computation is counted. (Chapter 8)

Brain training

Commercial games and exercises sold to improve the mind in general. They improve performance on the trained tasks, less on closely related ones, and show little evidence of improving everyday cognition. (Chapter 14)

Canonical microcircuit

The six-layered circuit repeated with only local variations across the neocortex: thalamic input arrives mainly in layer 4, layers 2 and 3 process and connect to other areas, layer 5 sends output,

and layer 6 feeds back to the thalamus. It is approximately the same whatever the cortex is processing. (Chapter 4)

Catastrophic interference
The failure, first shown in artificial networks, in which learning a new set of patterns erases previously learned ones. The brain is thought to avoid it by pairing a fast-learning hippocampus with slow, interleaved learning in the cortex. (Chapter 1)

Central pattern generator
A small network in the spinal cord or brainstem that produces a rhythmic motor pattern, such as walking or breathing, without rhythmic input. The minimal version, a half-center oscillator, is two mutually inhibiting populations that each fatigue while active; sensory feedback tunes the rhythm but does not drive it. (Chapter 4)

Cerebellum
A structure at the back of the brain holding about 69 of the brain's 86 billion neurons, specialized for the precision and timing of movement. Each Purkinje cell receives about 200,000 weak parallel fiber inputs and one powerful climbing fiber input that signals errors; by changing synapses where the two coincide, the cerebellum learns to predict and correct movements before errors occur. (Chapter 4)

Chain of thought
A method in which a language model writes out intermediate steps before answering, which helps it solve more multistep problems. Models trained by reinforcement learning to reason at length in this way have improved greatly on checkable problems in mathematics and programming. (Chapter 10)

Charles Bonnet syndrome
Vivid, detailed hallucinations in people who lose their sight late in life, while knowing that nothing is there. The book explains them as priors running unopposed once the sensory evidence that would correct them is gone. (Chapter 6)

Chunk
A unit formed by grouping several items or actions so that they are held in memory or executed as one. Experts recognize familiar configurations as single chunks, and practiced motor sequences come to be performed in groups that run off whole and cannot easily be revised in the middle. (Chapter 9)

Circuit
A group of interconnected neurons whose collective activity performs a specific transformation or regulatory function. (Chapter 4)

Cognition
Processes such as perception, memory, reasoning, and decision-making, treated here as emergent descriptions of neural dynamics.

Cognitive offloading
The use of physical action or external tools, such as writing a list or setting a reminder, to reduce the demands of a task. It usually improves performance on the task at hand but can reduce memory for what was offloaded. (Chapter 14)

Coincidence rule
A learning rule that strengthens only the synapses active at the moment a reward arrives. Because rewards come after the activity that earned them, it loses its selectivity once the reward is delayed beyond the cue, whereas a rule with an eligibility trace keeps crediting the right synapse. (Chapter 16)

Common currency
A single neural scale of value on which unlike options are priced before being compared, supported by value signals in the orbitofrontal cortex. Critics argue that choice could emerge from comparisons that never compute a single value. (Chapter 8)

Communication-through-coherence hypothesis
The proposal that brain regions route information by aligning their oscillations, so that one region's spikes arrive during the other's excitable phase, changing effective connectivity without changing any synapse. The evidence for it is mixed. (Chapter 4)

Complementary learning systems
The theory, set out by McClelland, McNaughton, and O'Reilly in 1995, that the brain avoids catastrophic interference by pairing a fast, sparse hippocampal learner that captures individual events with a slow cortical learner that extracts their statistical structure. The fast system trains the slow one by replaying its contents interleaved with existing knowledge. (Chapter 5)

Conductance-based synapse
A model synapse whose input arrives as a change in membrane conductance through particular receptor types, such as fast and slow excitatory receptors (including the NMDA receptor with its magnesium block) and inhibitory ones. (Chapter 16)

Constructionist view of emotion
The view, developed by Lisa Feldman Barrett, that there are no dedicated emotion programs; the brain continuously predicts and regulates the body, and a particular emotion is its categorization of that state in context, using concepts learned from experience and culture. It contrasts with the basic emotion view of a small set of evolved programs, each with its own bodily pattern and facial expression. (Chapter 7)

Context window
The text a language model holds in view during use, such as the current conversation. Anything learned there is lost when the conversation ends, because it is never written into the weights. (Chapter 10)

Context-dependent gating
A remedy for forgetting in which each task uses a fixed, random subset of a network's hidden units while the rest are silenced, inspired by the finding that different tasks strengthen synapses on

different dendritic branches. Combined with weight protection it preserves most of what was learned across many sequential tasks. (Chapter 11)

Contextual interference
The paradox that practicing several skills in a mixed or random order makes performance worse during practice but better at later tests and on new variations, compared with practicing each in a block. It belongs to a family of conditions, including spacing and reduced feedback, that make practice look worse but learning better. (Chapter 9)

Contingency
Responsiveness of a teacher or partner, moment to moment, to what a particular learner has just done. Experiments with infants and toddlers suggest that young children learn language from a person or a screen only when someone responds to them. (Chapter 16)

Convergence
A solution that engineers invented for their own reasons and only afterward recognized in the brain, such as experience replay, temporal-difference learning, and transformer attention. It is evidence that brain and machine, facing the same problem, found the same solution, not that the brain was copied. (Chapter 11)

Convergent evolution
The independent appearance of similar solutions in lineages that started from different points, such as cortex-like circuits in birds and mammals. The book treats it as a sign that a constraint, rather than shared history, is doing the work. (Chapter 13)

Core affect
The two dimensions on which emotional states fall: valence, how pleasant or unpleasant a state is, and arousal, how activated or calm. They are where the brain's interoceptive estimate of the body is most directly felt. (Chapter 7)

Corpus callosum
The bundle of fibers joining the two cerebral hemispheres. Most of its fibers are thinner than a micrometer, so signals between the hemispheres travel slowly. (Chapter 13)

Cortical folding
The buckling of the cortical sheet into folds. Across mammals the degree of folding is predicted by the cortex's surface area and thickness rather than its number of neurons, and it arises physically when an outer layer grows faster than what lies beneath it, not because a gene specifies each fold. (Chapter 13)

Credit assignment problem
The problem of deciding which of the many components that contributed to an outcome deserve the credit or the blame. In the brain it is made harder by delayed feedback, the absence of correct answers, and overlapping recurrent circuits. (Chapter 3)

Critical period
The strict form of a sensitive period: a window early in life when experience can reorganize sensory cortex and which, under normal conditions, cannot be reopened once it closes, as when closing one

eye of a kitten for a few weeks permanently removes that eye's cortical representation. It opens and closes under the control of inhibitory maturation and neuromodulators, and its closing makes adult perception stable and adult perceptual learning slow. (Chapter 6)

Curriculum learning
Training a network on examples ordered from simple to hard, by analogy with shaping in animal training. Its benefits have been mixed; large models use a coarser curriculum ordered by data quality rather than difficulty. (Chapter 11)

Dale's principle
The rule that a biological neuron is either excitatory or inhibitory toward all its targets, unlike a unit in most artificial networks, whose weights can be positive or negative. (Chapter 4)

Dark-room problem
The objection that a creature minimizing surprise should find a dark, quiet room and stay there. The reply, that creatures expect to be fed and warm, works only by writing their goals into the model as expectations. (Chapter 12)

Decision fatigue
The idea that making a long series of decisions, or exerting self-control, draws on a resource that runs down and degrades later choices. Large replications have not supported short-term depletion of willpower; what holds up is that many hours of demanding work shift people toward easier and quicker options. (Chapter 14)

Degeneracy
The ability of structurally different elements to perform the same function, as opposed to an engineer's redundancy of identical copies. It makes biological systems robust to unanticipated failures and gives evolution room to change one element while others keep the function alive. (Chapter 1)

Deliberate practice
Effortful, solitary work on specific weaknesses with feedback, which Ericsson and colleagues argued accounts for most of the difference between experts and others. Later meta-analyses found it explains a much smaller share of performance (about 12 percent overall), so the amount of practice alone does not explain who becomes expert. (Chapter 9)

Delta rule
The simplest error-correcting learning rule: the new estimate is the old one plus a fraction of the prediction error, the fraction being the learning rate. It underlies temporal-difference learning. (Chapter 12)

Dendrite
Branching extensions of a neuron specialized for receiving synaptic input. (Chapter 2)

Dendritic spike
A regenerative electrical event generated locally in a dendrite, such as an NMDA spike or a calcium spike, when enough nearby synapses are active together. It lets individual branches act as semi-independent compartments that detect and amplify coincident input. (Chapter 2)

Dendritic spine
The small protrusion from a dendrite that carries most excitatory synapses in the cortex. Spines enlarge when their synapses are potentiated, a structural signature of learning. (Chapter 2)

Diffusion
The net spread of molecules from regions of high concentration to low, driven by random thermal motion and costing no energy. Because diffusion time grows with the square of distance, it crosses a synapse in a fraction of a microsecond but would take years along a meter-long axon. (Chapter 0)

Disinhibition
Inhibition of an inhibitor: a target held silent by default is released when an input suppresses the neuron inhibiting it. Because the default is off, nothing happens unless the circuit is explicitly opened; the basal ganglia are built almost entirely from it. (Chapter 4)

Divisive normalization
A computation in which each neuron's response is divided by the pooled activity of its neighbors, keeping responses within range. Applied to value signals, it makes the value of each option relative to the total on offer. (Chapter 8)

Dopamine
A neuromodulator released by neurons of the ventral tegmental area and substantia nigra in the midbrain, which project to the striatum and prefrontal cortex. A large population of these neurons signals reward prediction error, dopamine gates plasticity in the striatum, and its loss causes Parkinson's disease. (Chapter 3)

Drift diffusion model
A model of two-choice decisions in which a decision variable drifts toward one of two bounds at a rate set by the strength of the evidence, jostled by noise; which bound it hits gives the choice and when gives the reaction time. It is the continuous version of Wald's sequential probability ratio test and is optimal for evidence of fixed strength. (Chapter 8)

Efference copy
A copy of each motor command sent to the sensory system so that the sensory consequences of the movement can be predicted and subtracted. It is why the world looks stable when the eyes move and why you cannot tickle yourself. (Chapter 6)

Efficient coding
The principle, proposed by Barlow, that sensory neurons are organized to remove redundancy and use their limited, noisy range of responses fully, for example by matching a response curve to how often each input occurs, as Laughlin found in the fly. A related retinal strategy, predictive coding, passes on only the part of a signal that neighboring points could not predict. (Chapter 12)

Eight statements
The book's model of how the brain learns, summarized as eight claims: the brain serves a body; it is built from poor parts; it predicts; it learns locally and keeps globally; it remembers on several clocks; it turns value into action and action into habit; it is grown, not specified; and it is a compromise and an inheritance. (Chapter 15)

Elastic weight consolidation

A method for reducing catastrophic forgetting in artificial networks by making the weights that mattered most for earlier tasks harder to change. Its authors linked it to the stabilization of dendritic spines after learning. (Chapter 9)

Electrochemical gradient

The combined push on an ion from its concentration gradient (the tendency to spread from where it is concentrated to where it is scarce) and its electrical gradient (attraction toward opposite charge). Unequal ion distributions across a membrane store this energy, and every neural signal runs on it. (Chapter 0)

Eligibility trace

A transient, decaying mark left at a recently active synapse, lasting seconds to minutes, during which a neuromodulatory signal can turn the activity into a lasting change. It lets a reward that arrives after the activity still reach the synapses responsible. (Chapter 3)

Engram

The physical trace of a memory: a sparse set of neurons, and the strengthened connections among them, that is both necessary and sufficient for a specific memory. Once a hypothetical term, it became an experimental object when the cells active during learning could be labeled and later reactivated or silenced. (Chapter 5)

Entropy

A measure of disorder, or more precisely of the number of microscopic arrangements compatible with what can be observed. The second law of thermodynamics says total entropy rises in any spontaneous process, so a cell must spend energy continuously to keep its own structure ordered, exporting entropy to its surroundings as heat and waste. (Chapter 0)

E-prop

A local learning rule for spiking recurrent networks that approximates the gradient by combining an eligibility trace at each synapse with a learning signal broadcast to many neurons. It is the brain's three-factor rule rederived from gradient descent and comes close to exact backpropagation through time on several tasks. (Chapter 11)

Evolutionary mismatch

A misfit between mechanisms tuned in ancestral environments and the inputs of a changed modern world, which can make mechanisms that once worked well produce harmful results. Such claims are easy to make and hard to test, because the ancestral environment cannot be observed directly. (Chapter 13)

Excitation

Neural processes that increase the likelihood that a neuron will fire an action potential. (Chapter 2)

Excitation-inhibition balance

The state of an awake cortex in which each excitatory input to a neuron is tracked by a nearly equal inhibitory input a few milliseconds later. It holds neurons near threshold, making circuits both fast and sensitive and keeping them stable. (Chapter 4)

Excitatory postsynaptic potential (EPSP)

A small, graded, short-lived depolarization of the receiving neuron, usually a fraction of a millivolt to a few millivolts, that moves it toward the threshold for firing. EPSPs are produced mostly by glutamate and fade over tens of milliseconds. (Chapter 2)

Expert iteration

A machine learning scheme in which a fast network is trained to reproduce the choices of a slow search, so that fast intuition is trained by slow deliberation, as in AlphaGo Zero. It is the engineering counterpart of the brain's handover from deliberation to habit. (Chapter 9)

Explore-exploit dilemma

The trade-off between choosing the option that currently looks best and sampling alternatives to learn their value, at the cost of rewards forgone. Its simplest form is the multi-armed bandit problem, a row of slot machines with fixed but unknown payoffs. (Chapter 8)

Extinction

The decline of a learned response when its cue is repeatedly presented without the outcome. It is new learning, a competing memory that the cue is now safe, rather than erasure, so the original fear can return in a new context, with time, or after an unsignaled shock. (Chapter 7)

Fear conditioning

A form of Pavlovian learning in which a neutral cue, such as a tone, is paired with a mild shock until the cue alone produces freezing, a rise in blood pressure, and stress hormones. The association is stored by LTP at the cue's synapses in the lateral amygdala. (Chapter 7)

Feedback inhibition

A motif in which a population's own output recruits interneurons that inhibit it, so the harder it fires the harder it is suppressed. It acts like a thermostat, is the main reason cortical activity does not explode, and naturally produces rhythms. (Chapter 4)

Feedforward inhibition

A motif in which an input excites a target neuron and, through an interneuron, inhibits the same target a millisecond or two later. Only inputs that arrive close together can drive the target, which turns it into a coincidence detector. (Chapter 4)

Flash-lag effect

The illusion that a moving object appears ahead of a flash presented at the same place, as if the brain projects motion forward to compensate for processing delays. Whether it reflects extrapolation or other mechanisms is still argued. (Chapter 12)

Forgetting curve

The steep and then more gradual loss of retention over time, first measured by Hermann Ebbinghaus in 1885 and close to a power law. Spaced retrieval, sleep, and emotional significance all slow it. (Chapter 5)

Formal and functional competence

Formal competence in language is knowing its words and rules; functional competence is using language to reason, plan, and refer to the world. In the human brain the two depend on different

networks, so a system trained only on language could master the first without the second. (Chapter 10)

Forward model
An internal model that predicts the sensory consequences of a motor command; the cerebellum is a principal site. (Chapter 6)

Framing effect
A change in choice when identical outcomes are described differently, for example as lives saved rather than lives lost, because the description moves the reference point. (Chapter 8)

Free-energy principle
Friston's proposal that any system that keeps itself within a limited set of states can be described as minimizing variational free energy, an upper bound on the improbability (surprise) of its sensations under its own model. This variational quantity is not the thermodynamic free energy of Chapter 0. Critics argue it is a flexible modeling framework that cannot itself be tested; only the process theories built under it can. (Chapter 12)

Generative replay
A remedy for catastrophic forgetting in which a network, instead of storing old examples, generates internal representations resembling those of earlier tasks and replays them, as the hippocampus reactivates patterns rather than replaying a recording. (Chapter 11)

Genomic bottleneck (genome bottleneck)
Anthony Zador's term for the constraint that a genome carries far too little information to list a large brain's connections: the human genome falls short of the human brain's wiring list by a factor of more than half a million. The genome must therefore specify compact, general rules for wiring, so a large brain is grown by rules and finished by learning, and evolution is pushed to build in structure that makes later learning fast. (Chapters 10 and 13)

Glia
The non-neuronal support cells of the nervous system, about as numerous as neurons in the human brain. They insulate axons, regulate the chemistry around synapses, and clear debris. (Chapter 0)

Goodhart's law
The principle that a measure which becomes a target ceases to be a good measure, seen in machine learning when optimization against a proxy reward eventually lowers true quality. (Chapter 11)

Grown, not specified
The model's seventh statement: the genome sets up cell types, gradients, and learning rules rather than a list of connections, and development and experience supply the rest, which is why early experience matters so much and why learning is not optional. (Chapter 15)

Growth mindset
The belief that intelligence can grow. Short interventions to instill it have produced small effects at best, mainly among lower-achieving students, and meta-analyses disagree about whether the average effect differs from zero. (Chapter 14)

Hallucination (in language models)
A fluent, confident invention produced by a language model asked a question it cannot answer. It is partly a product of training and evaluation that reward a guess over an admission of not knowing. (Chapter 10)

Hebb's rule (Hebbian learning)
The principle, stated by Donald Hebb in 1949, that a synapse strengthens when the cells on either side of it are active together ("cells that fire together wire together"). It is local, needing only information available at the synapse, but on its own it only strengthens and would drive every active synapse to its ceiling. (Chapter 5)

Hemispheric specialization
The concentration of particular functions, such as the rapid processing of speech sounds, in one cerebral hemisphere. The book presents the conjecture that large brains do this because signals passing between the hemispheres are too slow for time-critical computation. (Chapter 13)

Hippocampus
A structure deep in the temporal lobe that learns from a single exposure and later replays what it has learned to the cortex. Its internal loop runs from the entorhinal cortex through the dentate gyrus, CA3, and CA1. (Chapter 1)

Homeostasis
The maintenance of the body's internal variables, such as temperature, blood acidity, and glucose, near stable set points by negative feedback loops that detect deviations and act to reverse them. The term was coined by Walter Cannon in the 1920s. (Chapter 7)

Homeostatic plasticity
Slow self-regulation in which neurons monitor their own average activity over a day or more and scale their synapses up or down to keep firing within a working range. Synaptic scaling is its best-studied form. (Chapter 1)

Hopfield network
A network that stores memories in the connections among its units, so that a fragment presented as input settles into the nearest stored pattern, a model of pattern completion. The classical version holds only about 0.14 random patterns per unit; modern versions, which compare a cue with every stored pattern, have vastly greater capacity. (Chapter 11)

Hyperbolic discounting
Devaluing a delayed reward steeply for short delays and shallowly for long ones, roughly in proportion to one over one plus the delay times an impatience constant. Unlike exponential discounting, it causes preferences to reverse as a reward draws near. (Chapter 8)

Hypothalamic-pituitary-adrenal axis (HPA axis)
The slow arm of the stress response: a three-step hormonal relay from the hypothalamus through the pituitary gland to the adrenal cortex, which releases cortisol. Cortisol peaks tens of minutes after a stressor begins, mobilizes glucose, alters immune function and memory, and feeds back to shut down its own release. (Chapter 7)

Hypothalamus

A structure of about four grams at the base of the forebrain that senses blood temperature, glucose, and salt concentration and corrects them through the autonomic nervous system, the pituitary gland, and behavior such as hunger and thirst. It also contains circuits for attack, flight, mating, and feeding. (Chapter 7)

Illusion of fluency

The mistaken sense that easy, smooth study, such as rereading or grouped practice, is producing good learning. Because fluency feels like learning, learners often judge the less effective method to be the better one. (Chapter 14)

In-context learning

The ability of a trained language model to learn a new task from a handful of examples placed in its input, without any change to its weights. (Chapter 10)

Inductive bias

An assumption about the world built into a learner that lets it generalize from little data, such as the convolutional assumption that a feature means the same thing wherever it appears. On one view this, rather than its parts, is what the brain could lend to machines. (Chapter 11)

Inhibition

Neural processes that decrease the likelihood that a neuron will fire, essential for stability and selectivity. (Chapter 4)

Inhibitory postsynaptic potential (IPSP)

A graded voltage change produced by an inhibitory synapse, usually by opening chloride or potassium channels, that moves the membrane away from threshold or clamps it near rest and so makes firing less likely. (Chapter 2)

Instance theory

Logan's account of automaticity, in which each performance of a task stores an instance and the response comes from a race between the original procedure and retrieval of stored instances. As instances accumulate retrieval wins more often, so automaticity is the replacement of computing with remembering; the theory predicts the power law of practice. (Chapter 9)

Insula

A lobe of cortex folded deep inside the lateral fissure that is the main cortical target of interoceptive signals. Its posterior part is proposed to hold a primary map of the body's internal state, which is re-represented toward the front and combined with information about the environment, motivation, and social context. (Chapter 7)

Interoception

The sensing of the body's internal condition: heartbeat, breathing, stretch of the stomach and bladder, temperature, blood chemistry, inflammation, itch, and pain. Like other senses it is an inference shaped by expectation. (Chapter 7)

Inverse problem
The problem of inferring the causes of a sensory signal from the signal itself, such as recovering surfaces in the world from a two-dimensional retinal image. It is ill-posed, because many different states of the world produce the same evidence. (Chapter 6)

Inverted retina
The arrangement of the vertebrate retina, in which light must pass through layers of neurons and blood vessels to reach photoreceptors at the back, and the fibers leaving the eye create a blind spot. The independently evolved octopus eye lacks it, and the book uses it as an example of a feature explained by history rather than by constraint. (Chapter 13)

Inverted U
The relationship in which brain function, notably in the prefrontal cortex, improves with moderate levels of norepinephrine, dopamine, or cortisol and declines at the high levels released under acute stress, when control shifts toward amygdala-driven and habitual responses. (Chapter 7)

Ion channel
A membrane protein forming a pore that lets one kind of ion through and opens or closes in response to voltage or to a chemical. When channels open, ions run down their gradient and the membrane voltage changes within milliseconds; voltage-gated sodium and potassium channels generate the action potential. (Chapter 0)

Iowa gambling task
A card game with four decks, two paying more per card but losing money over time and two paying less but gaining, used to study choice under uncertainty. Patients with ventromedial prefrontal damage keep choosing the bad decks, a deficit later traced largely to a failure of reversal learning. (Chapter 7)

Ironies of automation
The observations by Lisanne Bainbridge that automating a process leaves operators the hardest tasks, asks them to monitor a system that does the job better than they do, and lets their skills decay through disuse, so that the most reliable automated systems may need the most operator training. (Chapter 14)

James-Lange theory
The theory, from William James in 1884 and Carl Lange, that a stimulus sets off bodily changes directly and the feeling of an emotion is the perception of those changes: we are afraid because we tremble, not the reverse. (Chapter 7)

Large language model
A very large network trained mostly to predict the next word of text, then further trained with rewards from human preferences or checked answers. It matches people in performance on many language tasks and partly in representation, but learns by a procedure no brain is known to use. (Chapter 10)

Latent cause
A hidden cause that the brain infers to explain a set of experiences. On the latent-cause account of Samuel Gershman, David Blei, and Yael Niv, a new context is learned as a new cause rather than by

rewriting the old one, so returning to the old context brings back the old learning, as when extinguished fears return. (Chapter 15)

Lateral inhibition
A motif in which neighboring neurons suppress one another, so the most strongly driven one wins. It sharpens contrast and enables winner-take-all selection; in the retina it emphasizes edges and produces the illusory fringes called Mach bands. (Chapter 4)

Lateral intraparietal area (LIP)
A region of parietal cortex that helps plan eye movements, whose neurons ramp up their firing as evidence accumulates during a perceptual decision and reach a common level just before the response. It is probably one of several areas that keep the running total of evidence. (Chapter 8)

Leaky integrate-and-fire neuron
The simplest model neuron that captures the essentials of spiking: it integrates its input, leaks charge back toward rest with a set time constant, fires when its voltage crosses a threshold, and is briefly refractory afterward. (Chapter 16)

Learning
The process by which experience produces lasting physical changes in neural structure or function that alter future behavior. (Chapter 1)

Learning progress
The rate at which predictions are improving, proposed as what human curiosity seeks, in contrast to artificial curiosity rewarded by sheer unpredictability, which can be captured by noise such as a television tuned to random channels. A learner should seek what it can learn. (Chapter 11)

Learning styles
The popular belief that individuals learn better when taught in their preferred sensory mode, such as auditory, visual, or kinesthetic. Reviews have found virtually no evidence that matching instruction to a preferred style helps. (Chapter 14)

Learns locally and keeps globally
The model's fourth statement: a synapse changes according to the activity on its two sides, but whether the change is kept depends on a third factor, a broadcast signal of surprise, reward, or importance carried by a neuromodulator and arriving while a trace of the recent activity remains. (Chapter 15)

Long-term depression (LTD)
A lasting decrease in synaptic strength, produced by weak or poorly timed input that causes a smaller, more prolonged calcium rise, which activates enzymes that pull receptors out of the synapse. (Chapter 5)

Long-term potentiation (LTP)
A lasting increase in the strength of a synapse produced by brief, intense, correlated activity, first found in the rabbit hippocampus in 1973. Physically it involves more transmitter release, more receptors, and an enlarged dendritic spine. (Chapter 2)

Loss aversion
Weighting a loss more heavily than an equal gain, by a factor typically somewhat below two. Patients with amygdala damage show sharply reduced loss aversion. (Chapter 8)

Loss of plasticity
The gradual loss, in standard networks trained on long sequences of tasks, of the ability to learn anything new, as more of their units become inactive or redundant. It was remedied by continually reinitializing a small fraction of the least-used units, a method called continual backpropagation. (Chapter 10)

Melatonin
The hormone whose nightly secretion signals biological night. Its timing shifts later around puberty, and bright light in the evening, such as from a tablet screen, suppresses and delays it. (Chapter 14)

Memory
The persistence of the changes that learning makes, and the capacity they give a system to behave differently because of past experience. In the brain it is held in altered tissue, not as stored records, and it is divided among several systems working on different clocks. (Chapter 1)

Metacognition
The capacity of one's confidence to track one's own accuracy, which varies between people and has been linked to gray matter in the anterior prefrontal cortex. (Chapter 8)

Mixture of experts
A model design that activates only a small fraction of its parameters for each input, letting the model grow without a proportional growth in computation. It echoes the brain's economy of sparse activity. (Chapter 10)

Model-based learning
Decision-making that learns how actions lead to outcomes and plans by simulating their consequences; it is flexible and computationally expensive and supports goal-directed action. (Chapter 8)

Model-free learning
Decision-making that caches the value of each action from experience through the reward prediction error; it is fast and cheap but cannot adjust to a change in an outcome's value until it has experienced the change. (Chapter 8)

Moravec's paradox
The observation that computers are far more easily made to perform like adults at abstract tasks, such as checkers or intelligence tests, than given the perception and mobility of a one-year-old. Moravec's explanation was that effortless skills such as seeing and grasping were refined over hundreds of millions of years of evolution. (Chapter 10)

Motion aftereffect

The illusory motion in the opposite direction seen on a still surface after looking at sustained motion, such as a waterfall. Neurons tuned to the adapted direction reduce their responses, tipping the balance toward the opposite direction when the motion stops. (Chapter 12)

Motor adaptation

Error-driven recalibration of a movement that already works, so that it stays accurate when the body or the world changes, as when reaches straighten against a pushing robot arm or throws come back onto target through prism glasses. It depends on the cerebellum and restores previous performance rather than adding new capability. (Chapter 9)

Musician's dystonia (focal dystonia)

A loss of voluntary control that appears only in extensively trained movements, such as a musician's playing, affecting about one in 100 professional musicians. One hypothesis is that the plasticity that enlarges finger maps in cortex, pushed too far, merges them until the fingers cannot move independently. (Chapter 9)

Myelin

An insulating sheath of fatty membrane, laid down in many layers by neighboring support cells, that wraps long-range axons and is interrupted at intervals by short bare gaps called nodes. It lets the impulse jump from node to node (saltatory conduction), reaching about 100 meters per second at lower energy cost. (Chapter 2)

N400

A negative brain wave peaking about 400 milliseconds after a word, larger for a word the preceding words made improbable and smaller for a probable one. It shows the brain's sensitivity to how expected each word is. (Chapter 12)

Neuromodulator

A chemical such as dopamine, norepinephrine, serotonin, or acetylcholine, released by small groups of neurons with widely branching axons, that changes how neurons fire and whether their synapses are permitted to change. It carries context rather than detailed content and acts over seconds to minutes. (Chapter 2)

Neuromorphic chip

Hardware that computes with spikes and keeps memory beside the processing units, as the brain does, and runs some tasks on a small fraction of the power of a conventional processor (examples include IBM's TrueNorth and Intel's Loihi). Such chips run trained networks well but have not yet been shown to train large models. (Chapter 10)

Neuron

A specialized cell that processes and transmits information using electrical and chemical signals. (Chapter 2)

Neurotransmitter

A chemical released from an axon terminal when calcium entry triggers vesicles to fuse with the membrane. It diffuses across the synaptic cleft and binds receptors on the receiving cell, opening ion channels; glutamate is the main excitatory transmitter. (Chapter 2)

NMDA receptor

A type of glutamate receptor whose channel is plugged by a magnesium ion at rest and unblocked by depolarization, so it passes current, including calcium, mainly when glutamate binding and depolarization coincide. It drives local dendritic spikes and the calcium signal that triggers lasting synaptic change. (Chapter 2)

Norepinephrine

A neuromodulator released almost entirely by the locus coeruleus, a small nucleus in the pons that projects to nearly the whole forebrain and the cerebellum. It signals surprise and arousal, sets the gain of cortical circuits, and on one account reports unexpected uncertainty, the sign that the world has changed. (Chapter 3)

Obstetric dilemma

The traditional view that human babies are born helpless because a bigger head could not pass through a pelvis narrow enough for efficient walking. The book treats it as contested by a rival account in which birth comes when the mother's metabolism reaches the limit of what she can sustain. (Chapter 13)

Outcome devaluation

A test that distinguishes goal-directed actions from habits: after an outcome is made undesirable, goal-directed responding falls, while responding that has become habitual through extensive training continues. (Chapter 8)

Pallium

The region of the forebrain corresponding to the cortex. In birds it lacks the cortex's layered sheet but can hold large numbers of small, densely packed neurons. (Chapter 13)

Pareto front

The set of designs that cannot be improved on one goal without becoming worse on another. No point on it is optimal in the everyday sense; each is a different compromise, and when each of several tasks is best served by one ideal design (an archetype), the front fills the region between those archetypes. (Chapter 13)

Pattern completion

The regeneration of a whole stored pattern of activity from a partial cue, carried out by densely interconnected recurrent excitatory neurons such as those of hippocampal area CA3. (Chapter 4)

Pattern separation

The process, attributed to the dentate gyrus of the hippocampus with its sparse, strongly inhibited granule cells, that makes similar inputs less similar so that related memories are stored distinctly. (Chapter 4)

Pavlovian controller

A decision system, older than planning and habit, that emits fixed responses to cues, such as approach to cues predicting reward and withdrawal from cues predicting harm, whether or not they serve the animal. It can override instrumental learning, the learning of actions by their consequences. (Chapter 8)

Perceptron

Rosenblatt's 1958 learning machine, a network of simple units whose connection weights are adjusted after mistakes. A single layer of such units cannot compute some simple functions, such as exclusive-or. (Chapter 10)

Plasticity

The capacity of neurons and synapses to change their strength, excitability, or structure with use, and the mechanisms that make those changes. It is the physical basis of learning. (Chapter 1)

Posterior (Bayes's rule)

The probability distribution an observer should act on after combining a prior with the likelihood, the probability of the current sensations under each possible state of the world. By Bayes's rule it is proportional to prior times likelihood, and it leans toward whichever of the two is more precise. (Chapter 6)

Power law of practice

The regularity that time per trial falls as a power function of the amount of practice, so that each doubling of practice buys about the same proportional gain. Averaging across people can manufacture the pattern, and individual curves are often better fitted by an exponential, but the shape of rapid early gains and slow late gains holds. (Chapter 9)

Practice testing

Studying by actively recalling material, as in short quizzes with feedback, rather than by rereading or highlighting. It is one of only two study techniques given the highest rating for strength of evidence, and classroom trials confirm its benefit. (Chapter 14)

Precision weighting

Weighting each source of information, whether a prior or a sensory cue, by its precision, that is, by how reliable it is. It explains how the senses are combined, and predictive coding proposes that attention is the gain applied to prediction errors according to their expected precision. (Chapter 6)

Prediction

The brain's use of prior experience to anticipate future sensory input or outcomes. (Chapter 12)

Prediction error (sensory)

In predictive coding, the difference between what a cortical level predicted and what it actually received, a whole pattern of errors about content passed between cortical layers. It is distinct from the single reward prediction error signaled by dopamine neurons (Chapter 3), which concerns value and not content. (Chapter 6)

Predictive coding

A theory of cortical processing in which each level of a hierarchy sends a prediction down to the level below, which returns only the prediction error. The higher level revises its hypothesis until the error is small, so what ascends is the part of the input that was not expected. (Chapter 6)

Prior

In Bayesian inference, a probability distribution encoding what is usually true before the current evidence arrives. In the brain a prior is learned structure, the statistics of a lifetime of input written into synaptic weights, and it can be revised by experience. (Chapter 6)

Prospect theory

Kahneman and Tversky's 1979 description of how people actually choose under risk: they evaluate outcomes as gains and losses relative to a reference point, are risk averse for gains and risk seeking for losses, weight losses more than equal gains, and overweight small probabilities while underweighting large ones. (Chapter 8)

Quantal release

The release of neurotransmitter in fixed packets, or quanta, each the contents of one vesicle, with each impulse releasing a random number of them, often one or none. It makes synaptic transmission intrinsically noisy and unreliable. (Chapter 2)

Receptive field

The region of the sensory field, for example the visual field, in which a stimulus can influence a neuron's firing. Receptive fields grow, and the stimuli that drive them become more abstract, at each stage of the visual hierarchy. (Chapter 6)

Reciprocal inhibition

The circuit arrangement in which a sensory input that excites one muscle's motor neurons also excites an inhibitory interneuron that suppresses the opposing muscle, so the two never fight. (Chapter 4)

Reconsolidation

The process by which a retrieved memory becomes labile again for some hours and must be restabilized to persist. During that window it can be strengthened, weakened, or altered, which is why memories drift with retelling. (Chapter 5)

Recurrent excitation

A motif in which excitatory neurons excite one another, so a pattern of activity can sustain itself after its input is gone. It underlies working memory and pattern completion and, without inhibition to contain it, becomes a seizure. (Chapter 4)

Reinforcement learning

The branch of machine learning in which an agent learns by acting and observing the consequences, from feedback that is delayed, noisy, and evaluative rather than a correct answer. Its temporal-difference algorithm closely matches the dopamine reward prediction error. (Chapter 1)

Repetition suppression

The decline in neural response to a stimulus each time it is repeated. On its own it is weak evidence for prediction, since fatigue could produce it, but experiments that vary expectation while holding repetition fixed reveal a separate expectation suppression. (Chapter 6)

Representation

A pattern of neural activity or connection strength that varies systematically with something in the world, the body, or the brain's own state, so that it can be described as standing for it. In this book the word describes what activity carries, not a stored copy that is looked up. (Chapter 1)

Resting potential

The steady voltage across a neuron's membrane at rest, about 70 millivolts negative inside relative to outside, produced by the ion gradients and by the membrane's greater leakiness to potassium. Every signal the neuron produces is a brief departure from it. (Chapter 2)

Retrieval-augmented model

A language model that keeps a large database of text outside its weights and, given a question, finds and reads the most similar passages before answering. Its knowledge can be corrected or extended without retraining, much as the hippocampal index can, but the database never teaches the weights anything. (Chapter 11)

Reward hacking

The failure in which an agent optimizing a specified reward finds unintended ways to maximize it, a central problem of AI safety. The book treats addiction as the biological counterpart. (Chapter 7)

Reward model

A second network, trained to predict which answers people prefer, whose score a language model is trained to maximize. Because it is only a proxy, pushing optimization against it too far makes true quality fall while the score keeps rising. (Chapter 11)

Reward prediction error

The difference between the reward received and the reward predicted: positive (a burst of dopamine firing) for an unexpected reward, zero for a fully predicted one, and negative (a dip below baseline) when a predicted reward fails to arrive. It is a single number about value, distinct from the sensory prediction error of Chapter 6. (Chapter 3)

Robust yet fragile

The property, described by Marie Csete and John Doyle, of complex regulated systems that are highly robust to the disturbances they were built for but vulnerable to rare ones they were not, because robustness in one place is paid for with fragility in another. (Chapter 13)

Saccade

A rapid jump of the eyes, made about three times a second, between fixations of a few hundred milliseconds; vision is suppressed during each jump. The book treats each saccade as a decision about where evidence is most needed. (Chapter 6)

Self-organization

The emergence of complex, apparently designed structure from simple local rules followed by many elements, with no central controller or blueprint. In brain wiring, chemical gradients lay down a coarse map of connections and correlated activity then sharpens it. (Chapter 0)

Self-supervised learning

Training without labels by asking a system to predict part of its input from the rest, such as the next word from the words before or a hidden patch of an image from the visible ones. Most capable systems of the 2020s were trained mostly this way, which resembles the brain's situation of few labels and an endless stream of input. (Chapter 10)

Sensitive period

A window of early life in which a circuit is unusually open to being shaped by experience, with largely permanent effects. Closing one eye of a kitten during this window costs that eye most of its territory in visual cortex, while the same closure in an adult does nothing. (Chapter 1)

Sensory adaptation

The continual rescaling of a sensory neuron's responses to the current statistics of its input, so that it keeps using its full range as the input becomes more or less variable. It is not fatigue but the revision of a prediction about what the next input will be like. (Chapter 12)

Sensory prediction error (in motor adaptation)

In motor learning, the difference between where the brain predicted a movement would go and where it was sensed to go. It drives motor adaptation, even when correcting it spoils the task the person is trying to perform. For the predictive-coding sense, see Prediction error (sensory). (Chapter 9)

Sentence embedding

A list of numbers that a pretrained model assigns to a sentence so that sentences with similar meanings get similar lists. (Chapter 16)

Serotonin

A neuromodulator released by the raphe nuclei along the midline of the brainstem. Its role is the least settled of the four; it has been tied to patience, aversion, behavioral inhibition, and the regulation of plasticity. (Chapter 3)

Sharp-wave ripples

Brief high-frequency events in the hippocampus, mostly during slow-wave sleep and quiet waking rest, in which place cells replay recent sequences of activity compressed roughly twentyfold. Disrupting them impairs memory of the replayed experience and prolonging them improves it. (Chapter 5)

Shortcut learning

The tendency of a network trained to reduce its error to use whatever feature of the training data reduces it, whether or not it is the feature a person would use, as when a pneumonia detector learned to recognize which hospital an X-ray came from. Such shortcuts fail when the data change. (Chapter 10)

Signal-to-noise ratio

The strength of a signal relative to the noise mixed into it. Adding more channels, vesicles, or detectors raises it roughly in proportion to their number, but the information carried grows only as its logarithm, so each additional bit costs more than the last (a law of diminishing returns). (Chapter 13)

Sliding threshold

The calcium level separating depression from potentiation, which shifts with a neuron's recent average activity: it rises after intense firing and falls after quiet periods. The more a neuron fires, the harder its inputs become to strengthen, which makes the neuron self-correcting. (Chapter 5)

Slow-wave sleep

The deep stage of sleep in which the cortex produces large, slow waves. It dominates the first half of the night and carries the hippocampal replay that drives consolidation. (Chapter 5)

Small-world network

A network in which most connections are short and form dense local clusters while a few long connections link the clusters, so any two nodes are only a few steps apart while wiring stays cheap. The brain's region-to-region wiring has this form, organized around highly connected hubs. (Chapter 4)

Sodium-potassium pump

A membrane protein that spends one ATP per cycle to push three sodium ions out of the cell and pull two potassium ions in, so potassium stays concentrated inside and sodium outside. It charges the membrane's "battery" slowly and continuously, at large energy cost. (Chapter 0)

Somatic marker hypothesis

Antonio Damasio's proposal that options which have led to good or bad outcomes come to evoke bodily states, the somatic markers, which bias choice before reasoning finishes. Later work has found little support for the claim that bodily feedback is needed for good choices. (Chapter 7)

Spacing

Spreading study or review sessions out over time rather than massing them together. The best gap between sessions grows with how long the memory must last but becomes a smaller fraction of that time, and a review helps most when the memory has faded but can still be retrieved. (Chapter 14)

Speed-accuracy trade-off

The fact that quick decisions make more errors and slow ones fewer. In the diffusion model it is set by the height of the bound, which people and monkeys raise or lower according to the cost of time and of errors. (Chapter 8)

Spike-timing-dependent plasticity (STDP)

A learning rule in which a synapse strengthens if the presynaptic spike precedes the postsynaptic spike by a few milliseconds to a few tens of milliseconds, and weakens if the order is reversed. Its logic is causal: an input that could have helped cause the spike is rewarded, and one that arrived too late is penalized. (Chapter 5)

Spontaneous recovery

The return of an earlier adaptation after brief training in the opposite direction, once errors stop being informative. It is explained by two learning processes running in parallel: a fast one that learns and forgets quickly and a slow one that learns slowly and holds on. (Chapter 9)

Stability

The ability of a neural system to maintain functional operation despite noise and perturbation.

Stability-plasticity dilemma

The conflict between learning quickly from new experience and protecting what is already known: if every fluctuation of activity changed the wiring, new learning would overwrite old as fast as it was acquired. (Chapter 3)

Stages of skill learning

Fitts and Posner's description of a skill moving from a cognitive stage (working out what to do, step by step), through an associative stage (linking the pieces as errors fall), to an autonomous stage (running with little attention). The stages describe the change but do not explain its mechanism. (Chapter 9)

Striatal spiral (ascending spiral)

The chain of connections by which each region of the striatum influences the dopamine neurons that project to the next region along, from the ventral striatum of motivation through the associative striatum to the sensorimotor striatum. It offers a route by which learning in one loop shapes what dopamine teaches in the next. (Chapter 9)

Stroop effect

The slowing of color naming when a word printed in colored ink names a different color, as when "red" is printed in green. It shows that reading has become automatic: the word is read whether or not reading is wanted. (Chapter 9)

Successor representation (predictive map)

A representation, from reinforcement learning, in which each location or state is encoded by the locations expected to follow it. Proposed as what the hippocampus encodes, it explains why place fields grow backward along a habitual route and unites the hippocampus's roles in memory and navigation. (Chapter 12)

Supernormal stimulus

An artificial object that triggers an instinctive response more strongly than the natural object the response evolved for, such as a giant egg that a bird tries to brood in preference to its own. (Chapter 13)

Surrogate gradient

A training trick for spiking networks that, during learning only, replaces the all-or-none spike's useless slope (zero except at threshold) with a smooth slope near threshold, so that backpropagation can be used. (Chapter 11)

Synapse

The junction at which one neuron influences another. At a chemical synapse an arriving impulse releases neurotransmitter across a gap about twenty nanometers wide onto receptors of the receiving cell; synapses both carry signals and are where the brain's memory is written. (Chapter 1)

Synaptic consolidation

The hour-scale stabilization of newly changed synapses, the transition from early to late LTP through protein synthesis and spine growth. Until it is complete, a memory can be blocked by drugs, a competing experience, or a blow to the head. (Chapter 5)

Synaptic scaling

A slow homeostatic adjustment, taking a day or two, in which every excitatory synapse on a neuron is scaled up or down by the same factor when its firing stays too low or too high. Because the change is multiplicative, the relative strengths that encode what was learned are preserved. (Chapter 5)

Synaptic tagging and capture

A molecular mechanism in which early long-term potentiation sets a tag at the synapse lasting an hour or two, and the tagged synapse captures newly made proteins that make the change lasting. It is a slower, hour-scale relative of the eligibility trace. (Chapter 5)

Systems consolidation

The gradual transfer, over weeks to years, of a memory from dependence on the hippocampus to independence from it, as repeated reactivation strengthens direct cortical connections. Rival accounts (multiple-trace and trace-transformation theory) hold that only a schematic version becomes independent, while the detailed episode stays hippocampal. (Chapter 5)

Task-bracketing

A pattern of striatal activity, seen with extended training, in which firing concentrates at the start and end of a learned action sequence with little in between. Graybiel proposed it is how the basal ganglia package a sequence of actions into a single unit. (Chapter 9)

Task-switching cost

The slowing and extra errors on the first trial after switching between tasks, reflecting the time the brain needs to reconfigure which rules are active. Warning of the switch in advance reduces the cost but does not eliminate it. (Chapter 14)

Temporal-difference learning

An algorithm for learning to predict future reward in which the learner compares successive predictions and uses their difference to correct the earlier one. Over trials the error migrates backward from the reward to the cue that predicts it, just as dopamine responses do. (Chapter 3)

Three-factor learning rule

The book's central learning rule. Activity on the two sides of a synapse (factors one and two) induces a change and sets an eligibility trace, and a broadcast neuromodulatory signal (the third factor) decides whether, how strongly, and in which direction the change is kept. (Chapter 3)

Threshold

The membrane voltage at which a neuron generates an action potential. (Chapter 2)

Tinbergen's four questions

The four kinds of answer, set out by the ethologist Niko Tinbergen in 1963, to the question of why an animal has a feature: its causation (the mechanism that produces it), its development in the individual, its survival value, and its evolutionary history. (Chapter 13)

Trade-off

A relationship in which improving a design on one goal costs it on another, so that no single design is best at both. The book argues that many of the brain's features are positions on trade-offs among

hard constraints of energy, space, time, noise, and genetic information, rather than optimal solutions to a single problem. (Chapter 13)

Transfer
The carryover of improvement from practiced tasks to untrained ones. A large online study of brain training found gains on the practiced tasks but no transfer to untrained tasks, even closely related ones; a later review found less improvement on closely related tasks and little on everyday cognition. (Chapter 14)

Transformer (attention)
The network architecture behind current language models. Its core operation, attention, weights each part of the input by its learned relevance to the current one, and one step of recall in a modern Hopfield network is mathematically the same operation, so the context window works as a one-shot associative memory. (Chapter 11)

Ultimatum game
An economic game in which one player proposes how to split a sum and the other accepts, so both are paid, or rejects, so neither is. Responders often reject unfair offers at a cost to themselves, a sense of fairness that varies across cultures. (Chapter 8)

Unconscious inference
Helmholtz's term for the perceptual system's automatic settling on the state of the world that would most probably have produced the current sensations. It is not made by reasoning and cannot be overruled by it, which is why an illusion persists after it is known to be one. (Chapter 6)

Utility
The value of an outcome to the chooser, as opposed to its amount, as proposed by Daniel Bernoulli in 1738. Expected utility theory holds that a chooser whose preferences obey a few consistency rules behaves as if maximizing utility weighted by probability. (Chapter 8)

Ventral stream
The visual pathway from the primary visual cortex (V1) through V2 and V4 to the inferotemporal cortex (IT), along which neurons respond to increasingly abstract and invariant features, ending in selectivity for objects and faces. A parallel dorsal stream specializes in motion and location. (Chapter 6)

Ventromedial prefrontal cortex (vmPFC)
A region of the prefrontal cortex that suppresses the amygdala's learned fear after extinction and links value with the body, receiving interoceptive signals and commanding the hypothalamus and brainstem. Damage to it leaves reasoning largely intact but impairs choice. (Chapter 7)

Volume transmission
Release of a neuromodulator into the extracellular space, often from swellings along the axon called varicosities, so that it diffuses over micrometers and reaches many cells and synapses at once, in contrast to point-to-point synaptic transmission. (Chapter 3)

Wanting and liking

The separation, proposed by Kent Berridge and Terry Robinson, of reward into wanting, the dopamine-dependent pull of a cue, and liking, the pleasure of the reward itself, which survives the loss of dopamine. Their divergence, especially in addiction, lets what an animal pursues come apart from what it enjoys. (Chapter 7)

White matter

The brain tissue made of wiring: the axons, many of them myelinated, that connect the gray matter of cell bodies and synapses. Across mammals its volume grows faster than that of gray matter as brains get larger. (Chapter 13)

Wiring economy (wire minimization)

The principle that, because wire costs space, energy, and conduction time, a nervous system should place its parts and route its connections to keep total wiring short. The brain does not minimize wire outright: it pays for some long-range connections to keep every part only a few steps from every other, a trade between wiring cost and short paths. (Chapter 13)